

Name: Tuba SamStudent-ID: CSC-25F-602Section : 2C (Evening)Subject : Linear AlgebraTeacher : Mr. Guresh Kumar

1. Solve the following system of linear equation the field of scalar being R :

$$2x_1 + x_3 = 1$$

$$2x_1 + 4x_2 - x_3 = -2$$

$$x_1 + 8x_2 - 3x_3 = 2$$

Sol.

making matrices form

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 4 & -1 \\ 1 & -8 & -3 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

Augmented Matrices

$$A_b = \left[\begin{array}{ccc|c} 2 & 0 & 1 & 1 \\ 2 & 4 & -1 & -2 \\ 1 & -8 & -3 & 2 \end{array} \right] \quad \text{Replace } R_1 \leftrightarrow R_3$$

$$A = \begin{bmatrix} 1 & -8 & -3 & 2 \\ 2 & 4 & -1 & -2 \\ 2 & 0 & 1 & 1 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - R_1 \end{array}$$

$$A = \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 20 & 5 & -6 \\ 0 & 16 & 7 & -3 \end{bmatrix} \quad \begin{array}{l} R_3 = R_3 - \frac{16}{20} R_2 \end{array}$$

$$A = \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 20 & 5 & -6 \\ 0 & 0 & 3 & \frac{39}{5} \end{bmatrix}$$

For

$$0 \quad 0 \quad 3 \quad \frac{39}{5}$$

Values of variables (Back Substitution)

$$x_1 - 8x_2 - 3x_3 = 2 \rightarrow (1)$$

$$20x_2 + 5x_3 = -6 \rightarrow (2)$$

$$3x_3 = \frac{9}{5} \rightarrow (3)$$

From eq (3):

$$x_3 = \frac{93}{5 \times 8} \Rightarrow x_3 = 0.6$$

put x_3 in eq (2)

$$20x_2 + 5(0.6) = -6$$

$$20x_2 + 3 = -6$$

$$20x_2 = -6 - 3$$

$$= -9$$

$$x_2 = -\frac{9}{20}$$

$$x_2 = -0.45$$

Bilal

put x_3 & x_2 in eq (1)

$$x_1 - 8(-0.45) - 3(0.6) = 2$$

$$x_1 + 1.8 = 2$$

$$x_1 = 2 - 1.8$$

$$x_1 = 0.2$$

So values are:

$$x_1 = 0.2, \quad x_2 = -0.45, \quad x_3 = 0.6$$

2:

$$x_1 + x_2 + x_3 = a$$

$$x_1 + (1+a)x_2 + x_3 = 2a$$

$$x_1 + x_2 + (1+a)x_3 = 3a$$

Sol.

Form into matrices:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{bmatrix} \quad \begin{array}{l} x_1 \\ x_2 \\ x_3 \end{array} = \begin{array}{l} a \\ 2a \\ 3a \end{array}$$

Augmented Matrices:

$$A_{\text{th}} = \begin{bmatrix} 1 & 1 & 1 & a \\ 1 & 1+a & 1 & 2a \\ 1 & 1 & 1+a & 3a \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - R_1 \end{array}$$

$$\begin{array}{cccc} -1 & -1 & -1 & -a \\ 0 & a & 0 & a \end{array}$$

$$A = \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & a & 0 & a \\ 0 & 0 & a & 2a \end{bmatrix}$$

$$\begin{array}{cccc} 1 & 1 & 1+a & 3a \\ -1 & -1 & -1 & -a \\ 0 & 0 & a & 2a \end{array}$$

$$\begin{array}{cccc} -1 & -1 & -1 & -a \\ 0 & 0 & a & 2a \end{array}$$

$$\begin{array}{cccc} 0 & 0 & a & 2a \end{array}$$

Bilal

Now for values of variables:

$$x_1 + x_2 + x_3 = a \rightarrow (1)$$

$$ax_2 = a \rightarrow (2)$$

$$ax_3 = 2a \rightarrow (3)$$

from eq (3)

$$ax_3 = 2a$$

$$x_3 = 2a/a$$

$$x_3 = 2$$

from eq (2)

$$ax_2 = a$$

$$x_2 = 1$$

Put x_3 & x_2 in eq (1)

$$x_1 + 1 + 2 = a$$

$$x_1 = a - 3$$

Final values are:

$$x_1 = a - 3$$

$$x_2 = 1$$

$$x_3 = 2$$

Ans.

$$3. \quad x_1 - x_2 + x_3 - x_4 + x_5 = 1$$

$$2x_1 + x_2 + 3x_3 + 4x_5 = 3$$

$$3x_1 - 2x_2 + 2x_3 + x_4 + x_5 = 1$$

$$x_2 + x_4 + x_5 = 0$$

Sol.

$$A = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 \\ 2 & 1 & 3 & 0 & 4 \\ 3 & -2 & 2 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{matrix} = \begin{matrix} 1 \\ 3 \\ 1 \\ 0 \end{matrix}$$

Augmented matrix

$$A_b = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 1 \\ 2 & 1 & 3 & 0 & 4 & 3 \\ 3 & -2 & 2 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{matrix} 1 \\ 3 \\ 1 \\ 0 \end{matrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 3 & 1 & 2 & 2 & 1 \\ 0 & 1 & -1 & 4 & -2 & -2 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1$$

$$R_2 \leftrightarrow R_4$$

$$R_3 = R_3 - 3R_1$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 4 & -2 & -2 \\ 0 & 3 & 1 & 2 & 2 & 1 \end{bmatrix}$$

$$R_3 = R_3 - R_2$$

$$R_4 = R_4 - 3R_2$$

$$R_3 = R_3 - R_2$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -3 & -2 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}$$

$$R_4 = R_3 + R_4$$

$$R_4 = R_4 - 3R_2$$

$$\begin{bmatrix} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -3 & -2 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{bmatrix}$$

Date _____

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 3 & -3 & -2 \\ 0 & 0 & 0 & 2 & -4 & -1 \end{array} \right]$$

Now for values of variables (Back Substitution)

$$x_1 - x_2 + x_3 - x_4 + x_5 = 1 \rightarrow (1)$$

$$x_2 + x_4 + x_5 = 0 \rightarrow (2)$$

$$-x_3 + 3x_4 - 3x_5 = -2 \rightarrow (3)$$

$$2x_4 - 4x_5 = -1 \rightarrow (4)$$

from eq (4)

$$2x_4 - 4x_5 = -1$$

$$2x_4 = -1 + 4x_5$$

$$x_4 = \frac{-1 + 4x_5}{2} \Rightarrow x_4 = -\frac{1}{2} + 2x_5$$

put x_4 in eq (3)

$$-x_3 + 3\left(\frac{-1 + 4x_5}{2}\right) - 3x_5 = -2$$

$$-x_3 - 3 + 12x_5 - 6x_5 = -2$$

$$-x_3 + 6x_5 = 1$$

$$1 + 6x_5 = x_3$$

$$0.5 + 3x_5 = x_3$$

put x_3 and x_4 in eq (2)

$$x_2 + \left(\frac{-1 + 4x_5}{2}\right) + x_5 = 0$$

$$2x_2 - 1 + 4x_5 + 2x_5 = 0$$

$$2x_2 - 1 + 6x_5 = 0$$

$$2x_2 + 6x_5 = 1$$

$$2(x_2 + 3x_5) = 1$$

$$x_2 + 3x_5 = \frac{1}{2}$$

$$x_2 = \frac{1}{2} - 3x_5$$

put x_2, x_3 and x_4 in eq (1)

$$x_1 - \left(\frac{1 - 3x_5}{2}\right) + \left(\frac{1 + 3x_5}{2}\right) - \left(\frac{-1 + 2x_5}{2}\right) + x_5 = 1$$

$$x_1 - \left(\frac{1 - 6x_5}{2}\right) + \left(\frac{1 + 6x_5}{2}\right) + \frac{1 - 2x_5}{2} + x_5 = 1$$

$$x_1 - 0.5 + 3x_5 + 0.5 + 3x_5 + 0.5 - 2x_5 + x_5 = 1$$

$$x_1 + 1.5 + 5x_5 = 1$$

$$x_1 + 5x_5 = 1 - 1.5$$

$$x_1 + 5x_5 = 0.5$$

$$x_1 = 0.5 - 5x_5$$

Final values:

$$x_1 = 0.5 - 5x_5, \quad x_2 = 0.5 - 3x_5, \quad x_3 = 0.5 + 3x_5,$$

$$x_4 = -0.5 + 2x_5$$

Date _____

$$\begin{aligned}
 4. \quad x_1 + x_2 - x_3 &= 1 \\
 x_2 + x_3 - x_4 &= 1 \\
 x_3 + x_4 - x_5 &= 1 \\
 -x_2 + x_4 + x_5 &= 1 \\
 -x_2 + x_3 + x_4 &= 1
 \end{aligned}$$

Sol.

form in matrix:

$$A = \begin{bmatrix} 1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 1 & 1 \\ 0 & -1 & 1 & 1 & 0 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{matrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

form Augmented Matrix:

$$A_b = \begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & -1 & 1 & 1 & 1 \\ 0 & -1 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{matrix} R_3 = R_3 + R_2 \\ R_4 = R_4 + R_2 \\ R_5 = R_5 + R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 0 & 0 & 2 \end{bmatrix} \begin{matrix} 0 & 0 & -1 & 1 & 1 & 1 \\ 0 & 0 & +1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ R_4 = R_4 + R_3 \\ R_5 = R_5 + R_4 \\ R_5 = R_5 - 2R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & -2 & 2 & 0 \end{bmatrix} \begin{matrix} R_5 - 2R_3 \\ 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & +2 & +2 & +2 & +2 \\ 0 & 0 & 0 & -2 & 2 & 0 \end{matrix}$$

Brilliant

$$\begin{bmatrix} 1 & 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 & 2 \end{bmatrix}$$

Now for values: (Back Substitution)

$$x_1 + x_2 - x_3 = 1 \rightarrow (1)$$

$$x_2 + x_3 - x_4 = 1 \rightarrow (2)$$

$$x_3 + x_4 - x_5 = 1 \rightarrow (3)$$

$$2x_4 = 2 \rightarrow (4)$$

$$2x_5 = 2 \rightarrow (5)$$

from eq (4)

$$2x_4 = 2$$

$$x_4 = 1$$

from eq (5)

$$2x_5 = 2$$

$$x_5 = 1$$

put x_4 & x_5 in eq (3)

$$x_3 + 1 - 1 = 1$$

$$x_3 = 1$$

put x_3 & x_4 in eq (2)

$$x_2 + 1 - 1 = 1$$

$$x_2 = 1$$

Brilliant

Date _____

put x_2 & x_3 in eq (1)

$$x_1 + \cancel{1} - \cancel{1} = 1$$

$$\boxed{x_1 = 1}$$

So, final values are:

$$x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 1, x_5 = 1.$$

* verification: put all values in eq (3)

$$x_3 + x_4 - x_5 = 1$$

$$1 + \cancel{1} - \cancel{1} = 1$$

$$1 = 1$$

Hence proved.

Ex.

$$x_1 - 2x_2 - 7x_3 + 7x_4 = 5$$

$$-x_1 + 2x_2 + 8x_3 - 5x_4 = -7$$

$$3x_1 - 4x_2 - 17x_3 + 13x_4 = 14$$

$$2x_1 - 2x_2 - 11x_3 + 8x_4 = 7$$

Solution:

form in matrix

$$A = \begin{bmatrix} 1 & -2 & -7 & 7 \\ -1 & 2 & 8 & -5 \\ 3 & -4 & -17 & 13 \\ 2 & -2 & -11 & 8 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} = \begin{matrix} 5 \\ -7 \\ 14 \\ 7 \end{matrix}$$

Date _____

Form Augmented Matrices:

$$A_b = \begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ -1 & 2 & 8 & -5 & | & -7 \\ 3 & -4 & -17 & 13 & | & 14 \\ 2 & -2 & -11 & 8 & | & 7 \end{bmatrix} \quad R_2 = R_2 + R_1$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ 0 & 0 & 1 & 2 & | & -2 \\ 3 & -4 & -17 & 13 & | & 14 \\ 2 & -2 & -11 & 8 & | & 7 \end{bmatrix} \quad \begin{matrix} R_3 = R_3 - 3R_1 \\ R_4 = R_4 - 2R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ 0 & 0 & 1 & 2 & | & -2 \\ 0 & 2 & 4 & -8 & | & -1 \\ 0 & 2 & 3 & -6 & | & -3 \end{bmatrix} \quad R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ 0 & 2 & 4 & -8 & | & -1 \\ 0 & 0 & 1 & 2 & | & -2 \\ 0 & 2 & 3 & -6 & | & -3 \end{bmatrix} \quad R_4 = R_4 - R_2$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ 0 & 2 & 4 & -8 & | & -1 \\ 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & -1 & 2 & | & -2 \end{bmatrix} \quad \begin{matrix} 0 & 0 & -1 & 2 & -2 \\ R_4 = R_4 + R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -2 & -7 & 7 & | & 5 \\ 0 & 2 & 4 & -8 & | & -1 \\ 0 & 0 & 1 & 2 & | & -2 \\ 0 & 0 & 0 & 4 & | & -4 \end{bmatrix}$$

Date _____

Form in equations for values:

$$x_1 - 2x_2 - 7x_3 + 7x_4 = 5 \rightarrow (1)$$

$$2x_2 + 4x_3 - 8x_4 = -1 \rightarrow (2)$$

$$x_3 + 2x_4 = -2 \rightarrow (3)$$

$$4x_4 = -4 \rightarrow (4)$$

from eq (4),

 $\Rightarrow$ put x_4 in eq (3)

$$4x_4 = -4$$

$$x_3 + 2(-1) = -2$$

$$x_4 = -1$$

$$x_3 - 2 = -2$$

$$x_3 = 0$$

Put x_4 & x_3 in eq (2)

$$2x_2 + 4(0) - 8(-1) = -1$$

$$2x_2 + 8 = -1$$

$$2x_2 = -9$$

$$x_2 = -\frac{9}{2}$$

$$2$$

$$x_2 = -4.5$$

put x_2, x_3 & x_4 in eq (1)

$$x_1 - 2(-4.5) - 7(0) + 7(-1) = 5$$

$$x_1 + 9 - 7 = 5$$

$$x_1 + 2 = 5$$

$$x_1 = 3$$

Final values:

$$x_1 = 3, x_2 = -4.5, x_3 = 0, x_4 = -1$$

B. I. Z. n

Ans.

Date _____

$$6. \quad x_1 + 2x_2 + x_3 = -1$$

$$6x_1 + x_2 + x_3 = -4$$

$$2x_1 - 3x_2 - x_3 = 0$$

$$x_1 - x_2 = 1$$

Sol.

Form in matrices

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & 1 & 1 \\ 2 & -3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} = \begin{matrix} -1 \\ -4 \\ 0 \\ 1 \end{matrix}$$

Form in augmented

$$A_b = \begin{bmatrix} 1 & 2 & 1 & -1 \\ 6 & 1 & 1 & -4 \\ 2 & -3 & -1 & 0 \\ 1 & -1 & 0 & 1 \end{bmatrix} \begin{matrix} R_2 = R_2 - 6R_1 \\ R_3 = R_3 - 2R_1 \\ R_4 = R_4 - R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & -11 & -5 & 2 \\ 0 & -7 & -3 & 2 \\ 0 & -3 & -1 & 2 \end{bmatrix} \begin{matrix} R_2 \leftrightarrow R_4 \\ R_2 = R_2 \times (-1) \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & -2 \\ 0 & -7 & -3 & 2 \\ 0 & -11 & -5 & 2 \end{bmatrix} \begin{matrix} R_3 = 3R_2 + 7R_2 \\ R_4 = 3R_4 + 11R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -4 & -16 \end{bmatrix} \begin{matrix} R_4 = R_4 - 2R_3 \end{matrix}$$

B. I. Z. n

Date _____

$$\begin{bmatrix} 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

from in equations

$$x_1 + 2x_2 + x_3 = -1 \rightarrow (1)$$

$$3x_2 + x_3 = -2 \rightarrow (2)$$

$$-2x_3 = -8 \rightarrow (3)$$

from eq, (3)

put x_3 in eq, (2)

$$+2x_3 = +8$$

$$3x_2 + 4 = -2$$

$$3x_2 = -2 - 4$$

$$3x_2 = -6$$

$$x_3 = 4$$

$$x_2 = -2$$

put x_3 & x_2 in eq, (1)

$$x_1 + 2(-2) + 4 = -1$$

$$x_1 - 4 + 4 = -1$$

$$x_1 = -1$$

So, final values :

$$x_1 = -1, \quad x_2 = -2, \quad x_3 = 4$$

Ans.

Brikan

Date _____

$$7. \quad 2x_1 + x_2 + 5x_3 = 4$$

$$3x_1 - 2x_2 + 2x_3 = 2$$

$$5x_1 - 8x_2 - 4x_3 = 1$$

Sol.

$$A_2 = \begin{bmatrix} 2 & 1 & 5 \\ 3 & -1 & 2 \\ 5 & -8 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$

$$A_6 = \begin{bmatrix} 2 & 1 & 5 & 4 \\ 3 & -1 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{bmatrix} \quad R_1 = R_1 \times \frac{1}{2}$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 3 & -1 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 5R_1 \end{array}$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 0 & -5/2 & -11/2 & -4 \\ 0 & -21/2 & -27/2 & -9 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 \times 2 \\ R_3 = R_3 \times 2 \end{array}$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 0 & 8 & 8 & 0 \\ 0 & 21 & 27 & 8 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 \times \frac{1}{8} \\ R_3 = R_3 \times \frac{1}{3} \end{array}$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 7 & 9 & 8/3 \end{bmatrix} \quad R_3 = 7R_2 - R_3$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & -8/3 \end{bmatrix} \quad R_3 = R_3 \times \left(-\frac{1}{2}\right)$$

$$= \begin{bmatrix} 1 & 1/2 & 5/2 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -5 & 0 \end{bmatrix} \quad R_3 \times \left(-\frac{1}{5}\right)$$

Brikan

Date _____

$$A = \left[\begin{array}{ccc|c} 1 & 1/2 & 5/2 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \quad R_2 = R_2 - R_3$$

$$A = \left[\begin{array}{ccc|c} 1 & 1/2 & 5/2 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$x_1 + \frac{1}{2}x_2 + \frac{5}{2}x_3 = 2 \rightarrow (1)$$

$$x_2 = 0 \rightarrow (2)$$

$$x_3 = 0 \rightarrow (3)$$

put x_2 & x_3 in eq (1)

$$x_1 = 2$$

Hence

$$x_1 = 2$$

$$x_2 = 0$$

$$x_3 = 0$$

Brilliant

Date _____

8. Solve the system of equations having the given matrices as their augmented matrices:

$$i) \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

Sol.

$$A_b = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right] \quad R_2 = R_2 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & -1 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right] \quad R_3 = R_3 + R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -1 & -1 & 2 \\ 0 & 0 & 0 & 3 \end{array} \right]$$

form in equations:

$$0x_1 + 0x_2 + 0x_3 = 3$$

$$0 = 3$$

solution is not possible, The system is inconsistent.

Brilliant

Date _____

$$\text{ii)} \begin{bmatrix} 1 & 2 & 4 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 0 \end{bmatrix}$$

Sol.

$$\Rightarrow \begin{bmatrix} 1 & 2 & 4 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 0 \end{bmatrix} \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - R_1 \\ R_4 = R_4 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \begin{array}{l} R_4 = R_4 + R_2 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & -4 & 0 \end{bmatrix} \begin{array}{l} R_4 = R_4 - \frac{4}{3} R_3 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 4 & 0 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

echelon form ban gai

for values of variables

$$\begin{array}{rcl} x_1 + 2x_2 + 4x_3 = 0 & \rightarrow \textcircled{1} \\ -x_2 - 3x_3 = 0 & \rightarrow \textcircled{2} \\ -3x_3 = 0 & \rightarrow \textcircled{3} \end{array}$$

Pr. I. n

Date _____

from eq ③

$$-3x_3 = 0$$

$$x_3 = 0$$

put x_3 in eq ②

$$-x_2 - 0 = 0$$

$$-x_2 = 0$$

$$x_2 = 0$$

put x_3 & x_2 in eq ①

$$x_1 + 2(0) + 4(0) = 0$$

$$x_1 = 0$$

final values :

$$x_1 = 0, x_2 = 0, x_3 = 0$$

Ans.

$$\text{iii)} \begin{bmatrix} 4 & 2 & -1 & 0 \\ 3 & 3 & 6 & 3 \\ 5 & 1 & -8 & 1 \end{bmatrix}$$

Sol.

$$\begin{bmatrix} 4 & 2 & -1 & 0 \\ 3 & 3 & 6 & 3 \\ 5 & 1 & -8 & 1 \end{bmatrix}$$

$$R_3 = 3R_2 \div 3$$

$$R_3 = 3R_2$$

$$R_3 = 1 \ 1 \ 2 \ 1$$

Pr. I. n

Date _____

$$\begin{bmatrix} 4 & 2 & -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \end{bmatrix} \quad R_3 = R_3 - 5R_2$$

$$\begin{bmatrix} 5 & 1 & -8 & 1 \end{bmatrix} \quad R_1 = R_1 - 4R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -9 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -4 & -18 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 & -8 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -5 & -10 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -4 & -18 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -9 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -9 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & -4 \end{bmatrix}$$

$$R_3 = R_3 - 2R_2$$

$$\begin{bmatrix} 0 & -4 & -18 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -4 & -18 & -8 \end{bmatrix}$$

$$\begin{bmatrix} + & + & + \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & -4 \end{bmatrix}$$

for values:

$$x_1 + x_2 + 2x_3 = 1 \rightarrow (1)$$

$$-2x_2 - 9x_3 = -4 \rightarrow (2)$$

$$0 = -4 \rightarrow (3)$$

from eq (2)

$$0 = -4$$

solution is not possible,
this is inconsistent.

Brilliant

Date _____

In each of the following use Gauss-Jordan method to reduce the given system to reduced echelon form, indicating the operations performed & determine the solution if any.

$$12. \quad 6x_1 - 6x_2 + 6x_3 = 6$$

$$2x_1 - 4x_2 - 6x_3 = 12$$

$$10x_1 - 5x_2 + 5x_3 = 30$$

Sol.

$$A = \begin{bmatrix} 6 & -6 & 6 \\ 2 & -4 & -6 \\ 10 & -5 & 5 \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 30 \end{bmatrix}$$

Augmented matrix:

$$A_b = \begin{bmatrix} 6 & -6 & 6 & 6 \\ 2 & -4 & -6 & 12 \\ 10 & -5 & 5 & 30 \end{bmatrix} \quad R_1 = R_1 \times \frac{1}{6}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 2 & -4 & -6 & 12 \\ 10 & -5 & 5 & 30 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1$$

$$R_3 = R_3 - 10R_1$$

$$\begin{bmatrix} 2 & -4 & -6 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -2 & -2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -2 & -8 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & -2 & -8 & 10 \\ 0 & 5 & -5 & 20 \end{bmatrix}$$

$$R_2 = R_2 \times \frac{1}{2}$$

$$R_3 = R_3 - 5R_2$$

$$\begin{bmatrix} 10 & -5 & 5 & 30 \end{bmatrix}$$

$$\begin{bmatrix} +10 & -10 & 10 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 5 & -5 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & -25 & 45 \end{bmatrix}$$

$$R_1 = R_1 + R_2$$

$$\begin{bmatrix} 0 & -1 & -4 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 5 & -5 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & -20 & 25 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & -25 & 45 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 5 & -4 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & -25 & 45 \end{bmatrix}$$

$$R_3 = R_3 \times \frac{1}{-25}$$

$$\begin{bmatrix} 1 & 0 & 5 & -4 \end{bmatrix}$$

Brilliant

Date _____

$$\begin{bmatrix} 1 & 0 & 5 & -4 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 1 & 9/5 \end{bmatrix}$$

$$R_2 = R_2 - 4R_3$$

$$R_1 = 5R_3 - R_1$$

$$\begin{bmatrix} 0 & 0 & 4 & 36/5 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 & 13 \\ 0 & -1 & 0 & 61/5 \\ 0 & 0 & 1 & 9/5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 4 & -5 \end{bmatrix}$$

$$\begin{bmatrix} - & - & - & + \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 0 & 61/5 \end{bmatrix}$$

form in equation

$$\begin{bmatrix} 0 & 0 & 5 & 9 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 5 & -4 \end{bmatrix}$$

$$\begin{bmatrix} - & - & - & + \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 0 & 13 \end{bmatrix}$$

$$-x_1 = 13$$

$$-x_2 = 61/5$$

$$x_3 = \frac{9}{5}$$

$$13. \quad 5x_1 + 5x_2 - x_3 = 0$$

$$10x_1 + 5x_2 + 2x_3 = 0$$

$$5x_1 + 15x_2 - 9x_3 = 0$$

Sol.

$$A = \begin{bmatrix} 5 & 5 & -1 \\ 10 & 5 & 2 \\ 5 & 15 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Ab = \begin{bmatrix} 5 & 5 & -1 & 0 \\ 10 & 5 & 2 & 0 \\ 5 & 15 & -9 & 0 \end{bmatrix}$$

$$R_1 = R_1 \times \frac{1}{5}$$

$$R_2 = R_2 - 10R_1$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & -5 & 4 & 0 \\ 5 & 15 & -9 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 5 & 2 & 0 \end{bmatrix}$$

$$R_3 = R_3 - 5R_1$$

$$\begin{bmatrix} -10 & -10 & -2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & 4 & 0 \end{bmatrix}$$

Date _____

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & -5 & 4 & 0 \\ 0 & 10 & -8 & 0 \end{bmatrix}$$

$$R_3 = R_2 + R_3$$

$$\begin{bmatrix} 5 & 15 & -9 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 5 & -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} - & - & + \end{bmatrix}$$

$$\begin{bmatrix} 0 & 10 & -8 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & -5 & 4 & 0 \\ 0 & 5 & -4 & 0 \end{bmatrix}$$

$$R_2 = R_2 + R_3$$

$$\begin{bmatrix} 0 & 10 & -8 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & +4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 5 & -4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 5 & -4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & 4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 5 & -4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & -5 & 4 & 0 \\ 0 & 10 & -8 & 0 \end{bmatrix}$$

$$R_2 = -\frac{1}{5} \times R_2$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & 1 & -4/5 & 0 \\ 0 & 10 & -8 & 0 \end{bmatrix}$$

$$R_3 = R_3 - 10R_2$$

$$\begin{bmatrix} 1 & 1 & -1/5 & 0 \\ 0 & 1 & -4/5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 = R_1 - R_2$$

$$\begin{bmatrix} -4/5 \times 10^2 \\ x_1 - 8 \\ + 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -3/5 & 0 \\ 0 & 1 & -4/5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{5} \times R_1$$

$$\begin{bmatrix} 0 & 1 & -4/5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & -1/5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -3/5 \end{bmatrix}$$

Brilliant

Brilliant

CS CamScanner

Date _____

$$x_1 + 0.6x_3 = 0 \quad \text{or } x_3 = t$$

$$x_1 = -\frac{3t}{5}$$

$$x_2 - \frac{4}{5}x_3 = 0$$

$$x_2 = \frac{4x_3}{5}$$

Ans.

Q. use the inversion algorithm to find the inverse of matrix (if it exists)

-12

$$a) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 2 & 5 & 3 & | & 0 & 1 & 0 \\ 1 & 0 & 8 & | & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & -2 & 5 & | & -1 & 0 & 1 \end{bmatrix} \begin{array}{l} R_3 = R_3 + 2R_2 \end{array}$$

$$\begin{array}{cc} -6 & -4 \\ -3 & 1 \\ -9 & -5 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_1 - R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & -3 & | & -2 & 1 & 0 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \begin{array}{l} R_2 = 3R_3 - R_2 \end{array}$$

Brilliant

$$R_1 = R_2 - R_1$$

Date _____

$$= \begin{bmatrix} 1 & 1 & 0 & | & -1 & 1 & 0 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_2 - R_1 \\ R_2 = 0 \quad 1 \quad 0 \quad 13 \quad -5 \quad -3 \end{array}$$

$$= \begin{bmatrix} -1 & 0 & 0 & | & 14 & -4 & -3 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & -1 & | & -5 & 2 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_1 \times -1 \\ R_3 = R_3 \times -1 \end{array}$$

$$= \begin{bmatrix} 1 & 0 & 0 & | & -14 & 4 & 3 \\ 0 & 1 & 0 & | & 13 & -5 & -3 \\ 0 & 0 & 1 & | & 5 & -2 & -1 \end{bmatrix}$$

So

$$A^{-1} = \begin{bmatrix} -14 & 4 & 3 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Ans.

$$b) \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 3 & -4 & | & 1 & 0 & 0 \\ 2 & 4 & 1 & | & 0 & 1 & 0 \\ -4 & 2 & 9 & | & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_1 \times -1 \\ R_2 = R_2 - 2R_1 \\ R_3 = R_3 + 4R_1 \end{array}$$

$$A^{-1} = \begin{bmatrix} 1 & -3 & 4 & | & -1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & -10 & 25 & | & -4 & 0 & 1 \end{bmatrix} \begin{array}{l} R_3 = R_2 + R_3 \\ R_2 = 2 \quad 4 \quad 1 \quad 0 \quad 1 \quad 0 \\ +2 \quad -6 \quad 8 \quad -2 \quad 0 \quad 0 \\ - \quad + \quad - \quad + \quad - \end{array}$$

$$A^{-1} = \begin{bmatrix} 1 & -3 & 4 & | & -1 & 0 & 0 \\ 0 & 10 & -7 & | & 2 & 1 & 0 \\ 0 & 0 & 18 & | & -4 & 0 & 1 \end{bmatrix} \begin{array}{l} 0 \quad 10 \quad -7 \quad 2 \quad 1 \quad 0 \\ 4 \quad -12 \quad 16 \quad -4 \quad 0 \quad 0 \\ -4 \quad 2 \quad 9 \quad 0 \quad 0 \quad 1 \\ 0 \quad -10 \quad 25 \quad -4 \quad 0 \quad 1 \end{array}$$

inverse not possible.

Brilliant

Date _____

$$12) \begin{array}{c} a.) \\ \left[\begin{array}{ccc|ccc} 1/5 & 1/5 & -2/5 & 1 & 0 & 0 \\ 4/5 & 1/5 & 1/10 & 0 & 1 & 0 \\ 1/5 & -4/5 & 1/10 & 0 & 0 & 1 \end{array} \right] \end{array}$$

Solution

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1/5 & 1/5 & -2/5 & 1 & 0 & 0 \\ 4/5 & 1/5 & 1/10 & 0 & 1 & 0 \\ 1/5 & -4/5 & 1/10 & 0 & 0 & 1 \end{array} \right] R_1 = R_1(5)$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 1/5 & 1/5 & 1/10 & 0 & 1 & 0 \\ 1/5 & -4/5 & 1/10 & 0 & 0 & 1 \end{array} \right] R_2 = R_2 - (1/5)R_1$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 0 & 1/2 & -1 & 1 & 0 \\ 0 & -1 & 1/2 & -1 & 0 & 1 \end{array} \right] R_2 \leftrightarrow R_3$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -1 & 1/2 & -1 & 0 & 1 \\ 0 & 0 & 1/2 & -1 & 1 & 0 \end{array} \right] \begin{array}{l} R_3 = R_3 \times (2) \\ R_2 = R_2(-1) \end{array}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 1 & -1/2 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array} \right] \begin{array}{l} R_1 = R_1 + 2R_3 \\ R_2 = R_2 + (1/2)R_3 \end{array}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 4 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array} \right] R_1 = R_1 - R_2$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 4 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array} \right]$$

Brilliant

Date _____

Hence

$$\text{inverse of matrix } A^{-1} = \left[\begin{array}{ccc} 1 & 4 & 0 \\ 0 & 0 & -1 \\ -2 & 2 & 0 \end{array} \right]$$

Ans.

In Exercise 3-6 use the following matrices to compute the indicated expression if it is defined.

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \quad E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

a.) $D + E$

$$\text{Sol.} \quad D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \quad E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$D + E = \begin{bmatrix} 7 & 6 & 5 \\ -2 & 1 & 3 \\ 7 & 3 & 7 \end{bmatrix}$$

b.) $D - E$

$$D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \quad E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

Brilliant

Date _____

$$D-E = \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

c) $5A$

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \Rightarrow 5A = \begin{bmatrix} 15 & 0 \\ -5 & 10 \\ 5 & 5 \end{bmatrix}$$

d) $-7C$

$$C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \Rightarrow -7C = \begin{bmatrix} -7 & -28 & -14 \\ -21 & -7 & -35 \end{bmatrix}$$

e) $2B-C$

Matrices can only be added subtracted

f) $4E-2D$

$$\text{Sol. } 4E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \Rightarrow 4E = \begin{bmatrix} 24 & 4 & 12 \\ -4 & 4 & 8 \\ 16 & 4 & 12 \end{bmatrix}$$

$$2D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \Rightarrow 2D = \begin{bmatrix} 2 & 10 & 4 \\ -2 & 0 & 2 \\ 6 & 4 & 8 \end{bmatrix}$$

$$4E-2D = \begin{bmatrix} 22 & -6 & 8 \\ -2 & 4 & 6 \\ 10 & 0 & 4 \end{bmatrix}$$

Ans.

Brihan

Date _____

$$\begin{aligned} 4.) \quad 5x_1 - 2x_2 + x_3 &= 3 \\ 3x_1 + 2x_2 + 7x_3 &= 5 \\ x_1 + x_2 + 3x_3 &= 2 \end{aligned}$$

Sol.

$$A = \begin{bmatrix} 5 & -2 & 1 \\ 3 & 2 & 7 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$$

$$A_b = \begin{bmatrix} 5 & -2 & 1 & 3 \\ 3 & 2 & 7 & 5 \\ 1 & 1 & 3 & 2 \end{bmatrix} R_1 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 3 & 2 & 7 & 5 \\ 5 & -2 & 1 & 3 \end{bmatrix} \begin{array}{l} R_2 = 3R_1 \\ R_3 = 5R_1 \end{array}$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -1 \\ 0 & -7 & -14 & -7 \end{bmatrix} R_2 = R_2 \times (-1)$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -7 & -14 & -7 \end{bmatrix} R_3 = R_3 + 7R_2$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_1 = R_1 - R_2$$

$$A_b = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 1-t, \quad x_2 = 1-2t, \quad x_3 = t$$

Ans.

Brihan

Q15

$$\begin{aligned} 5x_1 - 2x_2 + x_3 &= 2 \\ 3x_1 + 2x_2 + 7x_3 &= 3 \\ x_1 + x_2 + 3x_3 &= 2 \end{aligned}$$

Sol:

$$A = \begin{bmatrix} 5 & -2 & 1 \\ 3 & 2 & 7 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}$$

$$A_b = \begin{bmatrix} 5 & -2 & 1 & 2 \\ 3 & 2 & 7 & 3 \\ 1 & 1 & 3 & 2 \end{bmatrix} \quad R_1 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 3 & 2 & 7 & 3 \\ 5 & -2 & 1 & 2 \end{bmatrix} \quad \begin{aligned} R_2 &= R_2 - 3R_1 \\ R_3 &= R_3 - 5R_1 \end{aligned}$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & -7 & -14 & -8 \end{bmatrix} \quad R_2 \times (-1)$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & -7 & -14 & -8 \end{bmatrix} \quad R_3 = R_3 + 7R_2$$

$$= \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 13 \end{bmatrix} \quad R_1 = R_1 - R_2$$

$$= \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 13 \end{bmatrix}$$

$$0 = 13$$

solution is not possible

Brilliant

Q16

$$\begin{aligned} 2x_1 - x_2 + 3x_3 &= 3 \\ 3x_1 + x_2 - 5x_3 &= 0 \\ 4x_1 - x_2 + x_3 &= 3 \end{aligned}$$

Sol.

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 1 & -5 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$$

$$A_b = \begin{bmatrix} 2 & -1 & 3 & 3 \\ 3 & 1 & -5 & 0 \\ 4 & -1 & 1 & 3 \end{bmatrix} \quad R_2 = R_1 \times \left(\frac{1}{2}\right)$$

$$A_b = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 0 & -\frac{5}{2} & -\frac{11}{2} & -\frac{9}{2} \\ 0 & 1 & -5 & -3 \end{bmatrix} \quad \begin{aligned} R_2 &= R_2 - 3R_1 \\ R_3 &= R_3 - 4R_1 \end{aligned}$$

$$A_b = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 0 & 1 & -5 & -3 \\ 0 & \frac{5}{2} & -\frac{19}{2} & -\frac{9}{2} \end{bmatrix} \quad \begin{aligned} R_2 &\leftrightarrow R_3 \\ R_3 &= R_3 - \left(\frac{5}{2}\right)R_2 \end{aligned}$$

$$A_b = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 0 & 1 & -5 & -3 \\ 0 & 0 & 3 & 3 \end{bmatrix} \quad R_3 = R_3 \times \left(\frac{1}{3}\right)$$

$$= \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{aligned} R_1 &= R_1 + \left(\frac{1}{2}\right)R_2 \\ R_2 &= R_2 + 5R_3 \end{aligned}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad R_1 = R_1 + R_3$$

Brilliant

Date _____

Sol, $x_1 = 1$
 $x_2 = 2$
 $x_3 = 1$

Ans.

17.) $x_1 + 3x_2 + 5x_3 - 4x_4 + 0 = 1$
 $x_1 + 2x_2 + 1x_3 - x_4 + x_5 = -1$
 $x_1 - 2x_2 + 3x_3 + 2x_4 - x_5 = 3$
 $x_1 + 5x_2 + 3x_3 + x_4 + x_5 = -11$
 $x_1 + 3x_2 - 1x_3 + 2x_4 + x_5 = -3$

Sol

1	3	5	-4	0	x_1		1
1	2	1	-1	1	x_2		-1
1	-2	3	-2	-1	x_3	=	3
1	5	3	1	1	x_4		-11
1	3	1	1	1	x_5		-3

1	3	5	-4	0	1
1	2	1	-1	1	-1
1	-2	3	2	-1	3
1	5	3	1	1	-11
1	3	-6	1	1	-3

1	3	5	-4	0	1
0	-1	-4	3	1	-2
0	-5	-2	6	-1	-2
0	2	-2	5	1	-12
0	0	-6	5	2	-4

$R_2 = R_2 - R_1$
 $R_3 = R_3 - R_1$
 $R_4 = R_4 - R_1$
 $R_5 = R_5 - R_1$

Brilliant

Date _____

1	0	-7	5	6	-5
0	1	4	-3	-1	2
0	0	-18	-9	-6	12
0	0	-10	11	3	-6
0	0	-6	5	2	-4

$R_1 = R_1 - 3R_2$
 $R_2 = (-R_2)$
 $R_3 = R_3 - 5R_2$
 $R_4 = R_4 - 3R_2$

1	0	0	0	0	1
0	1	0	0	10	-1
0	0	1	0	0	1
0	0	0	1	0	-1
0	0	0	0	1	0

$R_1 = R_1 - 4R_3$
 $R_2 = R_2 - 4R_3$
 $R_3 = 10R_3$
 $R_4 = R_4 + 10R_3$
 $R_5 = R_5 + 6R_3$

$x_1 = 1$
 $x_2 = -1$
 $x_3 = 1$
 $x_4 = -1$
 $x_5 = 0$

Brilliant

$$\begin{bmatrix} 3 & 2 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ 3 \end{bmatrix}$$

Date _____

$$\begin{bmatrix} 3 & 2 & 4 & : & 7 \\ 2 & 1 & 1 & : & 4 \\ 1 & 3 & 5 & : & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 5 & : & 3 \\ 2 & 1 & 1 & : & 4 \\ 3 & 2 & 4 & : & 7 \end{bmatrix} \quad R_{12} \text{ (Row swapping)}$$

$$\begin{bmatrix} 1 & 3 & 5 & : & 3 \\ 0 & -5 & -9 & : & -2 \\ 0 & -7 & -11 & : & -2 \end{bmatrix} \quad \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 5 & : & 3 \\ 0 & 1 & 9/5 & : & 2/5 \\ 0 & -7 & -11 & : & -2 \end{bmatrix} \quad R_2 = (-1/5)R_2$$

$$\begin{bmatrix} 1 & 3 & 5 & : & 3 \\ 0 & 1 & 9/5 & : & 2/5 \\ 0 & 0 & 8/5 & : & 4/5 \end{bmatrix} \quad R_3 = R_3 + 7R_2$$

$$\begin{bmatrix} 1 & 3 & 2/5 & : & 9/5 \\ 0 & 1 & 9/5 & : & 2/5 \\ 0 & 0 & 1 & : & 1/5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & -1/2 \\ 0 & 0 & 1 & : & 1/2 \end{bmatrix} \quad \begin{array}{l} R_1 = R_1 + 2R_3 \\ R_2 = R_2 + 9/5 R_3 \\ R_3 = \frac{5}{2} R_3 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & -1/2 \\ 0 & 0 & 1 & : & 1/2 \end{bmatrix}$$

$$\begin{array}{l} m_1 = 2 \\ m_2 = -1/2 \\ m_3 = 1/2 \end{array}$$

Brilliant

$$\begin{array}{rcl} 5m_1 + & +4m_3 + 2m_4 & = 3 \\ m_1 - & m_2 + 2m_3 + m_4 & = 1 \\ 4m_1 + & m_2 + 2m_3 + 0 & = 1 \\ m_1 + & m_2 + m_3 + m_4 & = 0 \end{array}$$

Date _____

$$A_2 = \begin{bmatrix} 5 & 0 & 4 & 2 \\ 1 & -1 & 2 & 1 \\ 4 & 1 & 2 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 5 & 0 & 4 & 2 & : & 3 \\ 1 & -1 & 2 & 1 & : & 1 \\ 4 & 1 & 2 & 0 & : & 1 \\ 1 & 1 & 1 & 1 & : & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 1 & : & 1 \\ 5 & 1 & 4 & 2 & : & 3 \\ 4 & 1 & 2 & 0 & : & 1 \\ 1 & 1 & 1 & 1 & : & 0 \end{bmatrix} \quad R_1 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 1 & : & 1 \\ 0 & 1 & -6 & -3 & : & -2 \\ 0 & 1 & -6 & -4 & : & -3 \\ 0 & 1 & -1 & 0 & : & -1 \end{bmatrix} \quad \begin{array}{l} R_2 = 5R_1 \\ R_3 = R_4 - R_1 \\ R_4 = R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 2 & 1 & : & 1 \\ 0 & 5 & -6 & -3 & : & -2 \\ 0 & 5 & -6 & -4 & : & -3 \\ 0 & 2 & -1 & 0 & : & -1 \end{bmatrix} \quad R_2 = R_2 - R_3$$

$$\sim \begin{bmatrix} 1 & 1 & -4/5 & 2/5 & : & 3/5 \\ 0 & 1 & -4/5 & -3/5 & : & -7/5 \\ 0 & 5 & 0 & -2 & : & -1 \\ 0 & 2 & 7/5 & 6/5 & : & -1/5 \end{bmatrix} \quad \begin{array}{l} R_1 = R_1 + R_2 \\ R_3 = R_3 - 5R_2 \\ R_4 = R_4 - 2R_2 \end{array}$$

Brilliant

Date _____

$$\sim \begin{bmatrix} 1 & 0 & 4/5 & 2/5 & 3/5 \\ 0 & 1 & -6/5 & -3/5 & -2/5 \\ 0 & 0 & 1 & 4/7 & -1/7 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{array}{l} R_2 \left(\frac{5}{7} \right) R_3 \\ R_4 (-1) R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -2/7 & 5/7 \\ 0 & 1 & 0 & 3/7 & -4/7 \\ 0 & 0 & 1 & 4/7 & -1/7 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_1 - \left(\frac{4}{7} \right) R_3 \\ R_2 = R_2 + \left(\frac{4}{7} \right) R_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{array}{l} R_1 = R_1 + \left(\frac{2}{7} \right) R_4 \\ R_2 = R_2 - \left(\frac{3}{7} \right) R_4 \\ R_3 = R_3 - \left(\frac{4}{7} \right) R_4 \end{array}$$

$$x_1 = 1, \quad x_2 = -1, \quad x_3 = -1, \quad x_4 = 1$$

Brilliant

Q208

$$\begin{array}{rcl} 2x_1 - x_2 + 3x_3 & = & a \\ 3x_1 + x_2 - 5x_3 & = & b \\ -5x_1 - 5x_2 + 21x_3 & = & c \end{array}$$

is an inconsistent
if $c \neq 2a - 3b$

Date _____

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & -1 & -5 \\ -5 & -5 & 21 \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 3 & a \\ 3 & -1 & -5 & b \\ -5 & -5 & 21 & c \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 & a/2 \\ 0 & 5/2 & -19/2 & b-3a/2 \\ 0 & -5 & 21 & c \end{bmatrix} \begin{array}{l} R_1 = R_1/2 \\ R_2 = R_2 - 3R_1 \\ R_3 = R_3 + 5R_1 \end{array}$$

$$\begin{bmatrix} 1 & 1/2 & 3/2 & a/2 \\ 0 & 1 & -19/2 & \frac{2b-3a}{5} \\ 0 & -5/2 & 57/2 & \frac{c+5a}{2} \end{bmatrix} \quad R_3 = \left(\frac{2}{5} \right) R_2$$

$$\begin{bmatrix} 1 & -1/2 & 3/2 & a/2 \\ 0 & 1 & -19/2 & \frac{2b-3a}{5} \\ 0 & 0 & 0 & c-2a+3b \end{bmatrix}$$

So,

then

$$0 + 0 + 0 = c - 2a + 3b$$

$$0 = c - 2a + 3b$$

$$0 + c = -2a + 3b$$

$$c \neq -2a + 3b$$

consistent system } para.

Brilliant